

Polynomials, factor and remainder theorems - Set 2 - Solutions

Specification MM1.6.

Answer key

Question	Answer	Topic
1	H	Polynomials, factor and remainder theorems
2	B	Polynomials, factor and remainder theorems
3	A	Polynomials, factor and remainder theorems
4	B	Polynomials, factor and remainder theorems
5	F	Polynomials, factor and remainder theorems

Worked solutions

Question 1 POL-08 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following statements about the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is/are true?

I $p(x)$ has three distinct real roots.

II $(x - 4)$ is a factor of $p(x) - 6$.

III $p(x) > 0$ for all $x > 3$.

Answer: H

$p(x) = (x - 1)(x - 2)(x - 3)$, so I is true. $p(4) - 6 = 3 \cdot 2 \cdot 1 - 6 = 0$, so II is true. For $x > 3$ all three factors are positive, so III is true.

Question 2 POL-09 · Polynomials, factor and remainder theorems · spec MM1.6

The expression $x^4 + 4$ can be factorised into two quadratics with integer coefficients. What is the sum of the coefficients of x in these two quadratics?

Answer: B

$x^4 + 4 = (x^2 + 2)^2 - 4x^2 = (x^2 + 2x + 2)(x^2 - 2x + 2)$. The x -coefficients are 2 and -2 , summing to 0.

Question 3 POL-10 · Polynomials, factor and remainder theorems · spec MM1.6

Given that $x^3 + px + q$ is divisible by $(x - 2)^2$, find the value of q .

Answer: A

$x^3 + px + q = (x - 2)^2(x - r)$. Comparing x^2 coefficients: $-4 - r = 0 \Rightarrow r = -4$. Constant term: $(4)(4) = 16 = q$. (Also $p = 4 + 4r = -12$.)

Question 4 POL-12 · Polynomials, factor and remainder theorems · spec MM1.6

When $x^3 - 3x^2 + 2x + 7$ is divided by $x^2 + 1$, what is the remainder?

Answer: B

$x^3 - 3x^2 + 2x + 7 = (x^2 + 1)(x - 3) + (x + 10)$: expanding gives $x^3 - 3x^2 + x - 3 + x + 10$. Remainder $x + 10$.

Question 5 POL-16 · Polynomials, factor and remainder theorems · spec MM1.6

For which value of k does $x^2 - x + k$ divide exactly into $x^3 - 2x^2 + 3x - 4$? (Assume such a value exists.)

Answer: F

Try $x^3 - 2x^2 + 3x - 4 = (x^2 - x + k)(x + m)$: comparing x^2 : $m - 1 = -2 \Rightarrow m = -1$. Then x : $k - m = k + 1 = 3 \Rightarrow k = 2$. Constant: $km = -2 \neq -4$. No value of k works.